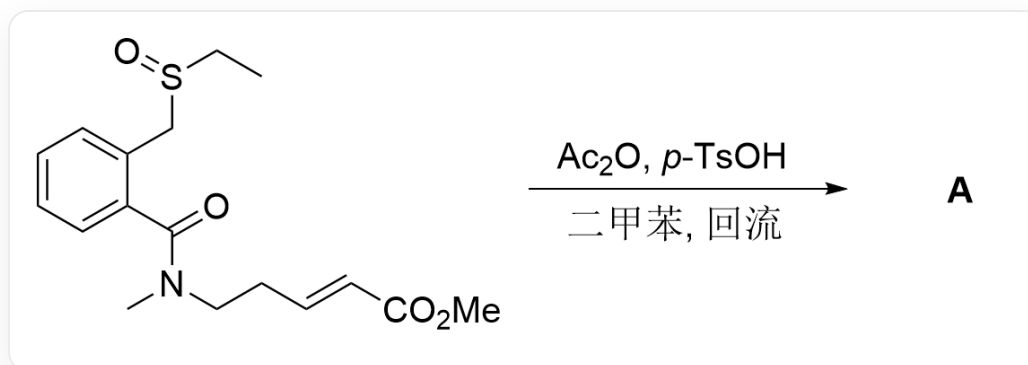


题目

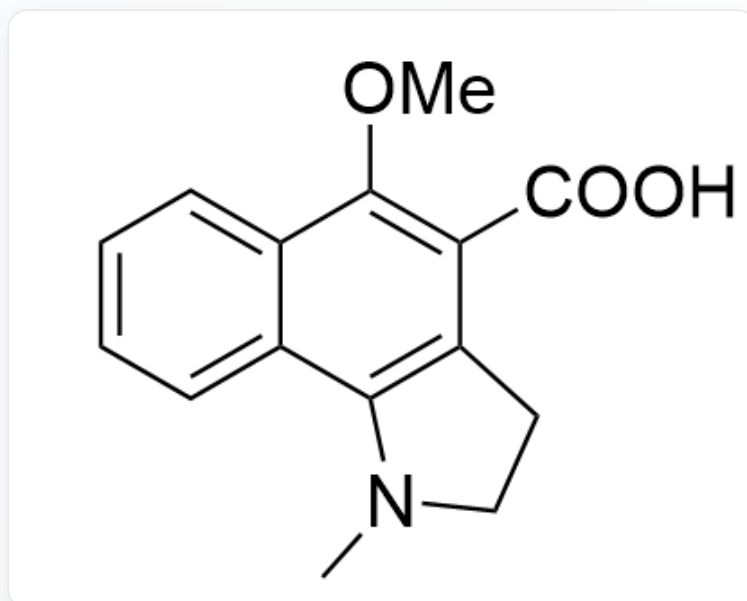


CCS(CC1=CC=CC=C1C(N(CC/C=C/C(OC)=O)C)=O)=O>CC1=CC=C(S(=O)(=O)C=C1.CC(OC(C)=O)=O,二甲苯,回流>[A],A是产物

已知反应产物 **A** 的分子式为 $\text{C}_{15}\text{H}_{15}\text{NO}_3$ ，**A** 中含有三个环，试给出 **A** 的结构式

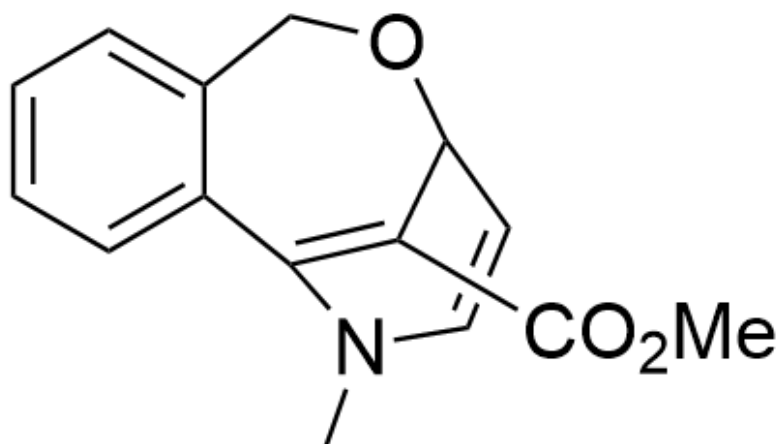
A. 其他选项均不正确

B.



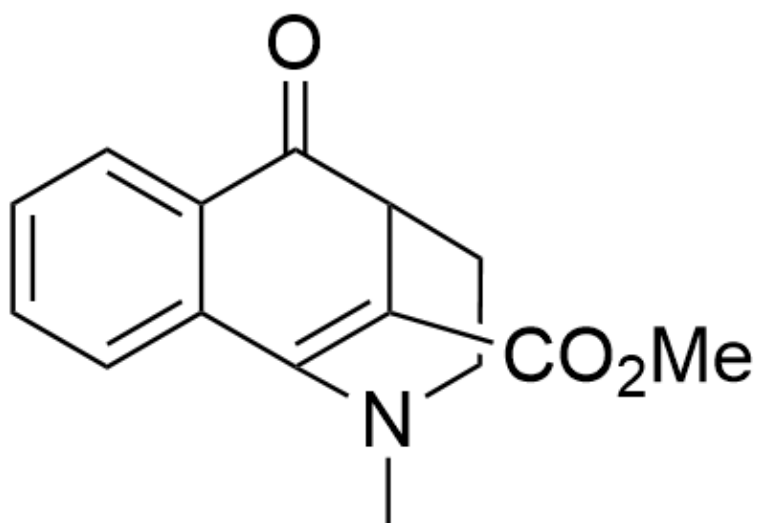
CN(CC1)C2=C1C(C(O)=O)=C(OC)C3=CC=CC=C32

C.



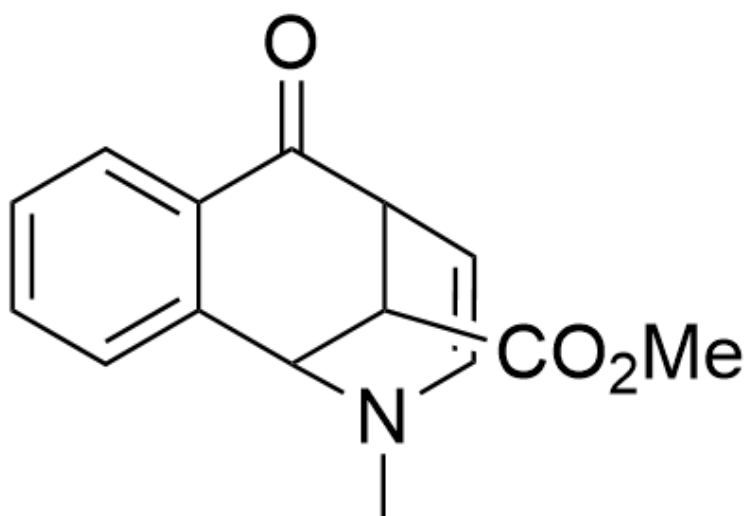
CN1C(C2=CC=CC=C2CO3)=C(C(OC)=O)C3C=C1

D.



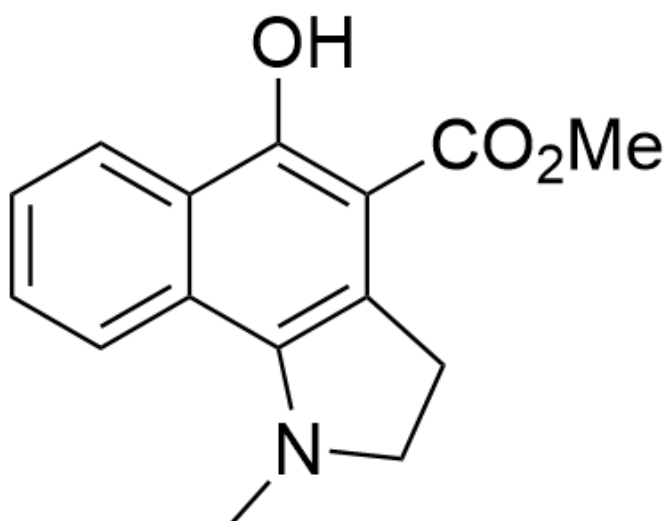
CN1C(C2=CC=CC=C2C3=O)=C(C(OC)=O)C3CC1

E.



CN(C=CC1C2C(OC)=O)C2C3=CC=CC=C3C1=O

F.



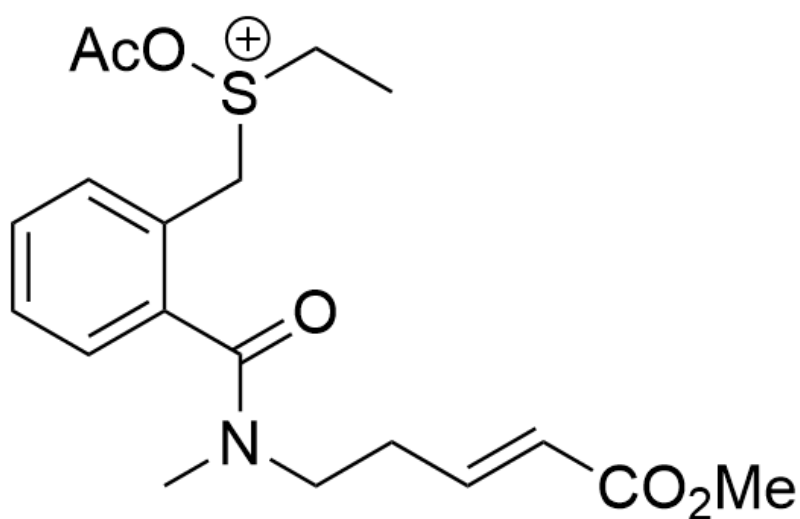
OC1=C(C(OC)=O)C2=C(N(C)CC2)C3=CC=CC=C31

答案

正确答案: **F**

详细解析

首先，亚砷中的氧原子被乙酰化，得到中间体 **1**



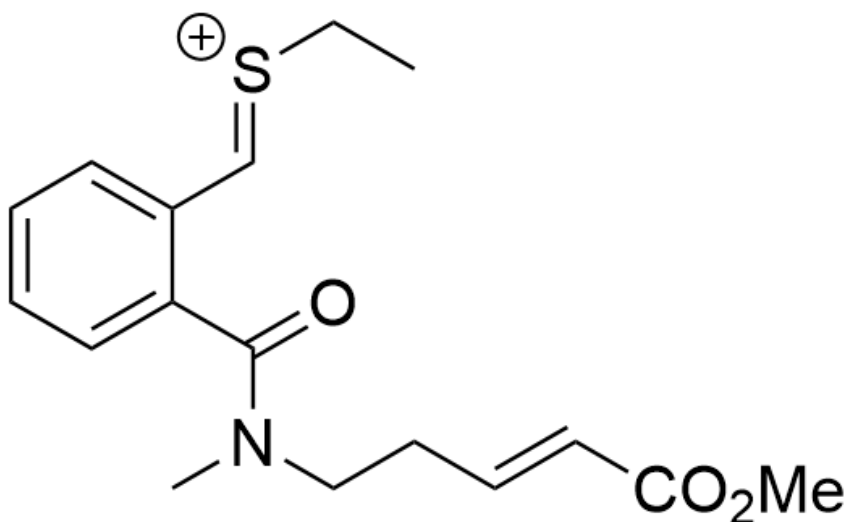
中间体 **1** : CC[S+](OC(C)=O)CC1=CC=CC=C1C(N(C)CC/C=C/C(OC)=O)C=O

CHECKPOINT

1 PTS

中间体 **1** : CC[S+](OC(C)=O)CC1=CC=CC=C1C(N(C)CC/C=C/C(OC)=O)C=O

接着，消除一分子乙酸得到反应中间体 **2**



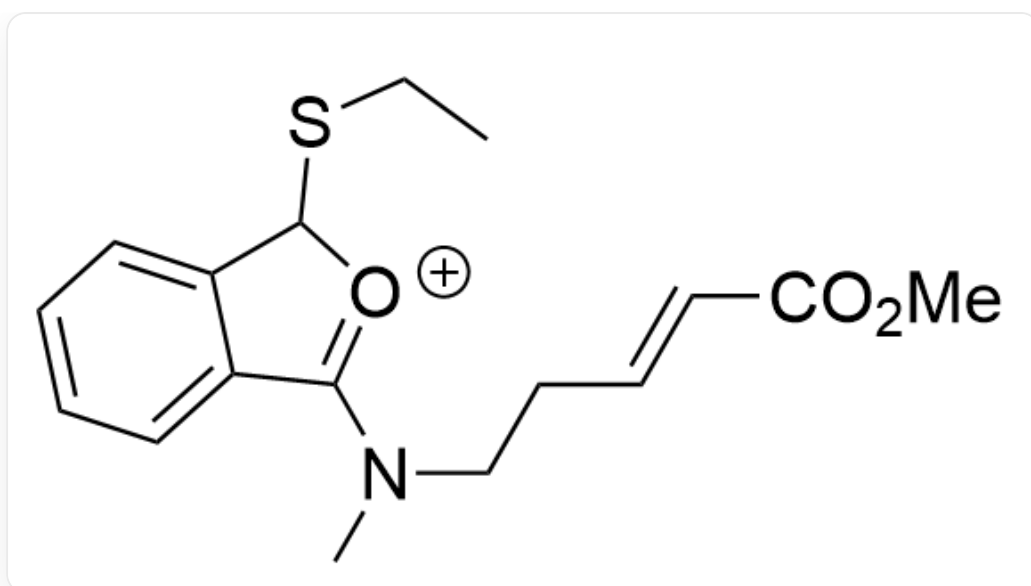
中间体2: CC/[S+]=C/C1=CC=CC=C1C(N(CC/C=C/C(OC)=O)C)=O

CHECKPOINT

1 PTS

中间体2: CC/[S+]=C/C1=CC=CC=C1C(N(CC/C=C/C(OC)=O)C)=O

临近的氧原子亲核进攻碳正离子形成五元环，得到中间体3



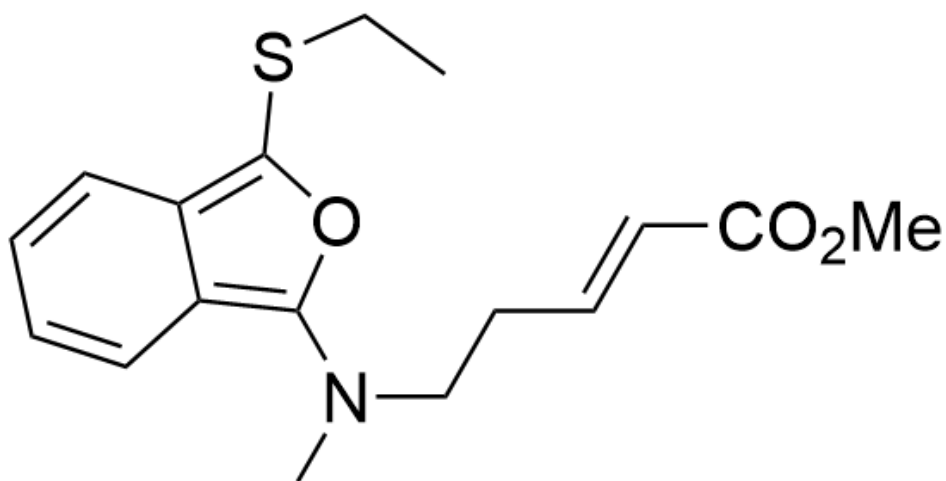
中间体**3**: CCSC1C2=CC=CC=C2C(N(C)CC/C=C/C(OC)=O)C1=[O+]

CHECKPOINT

1 PTS

中间体**3**: CCSC1C2=CC=CC=C2C(N(C)CC/C=C/C(OC)=O)C1=[O+]

接着去质子芳构化得到中间体**4**



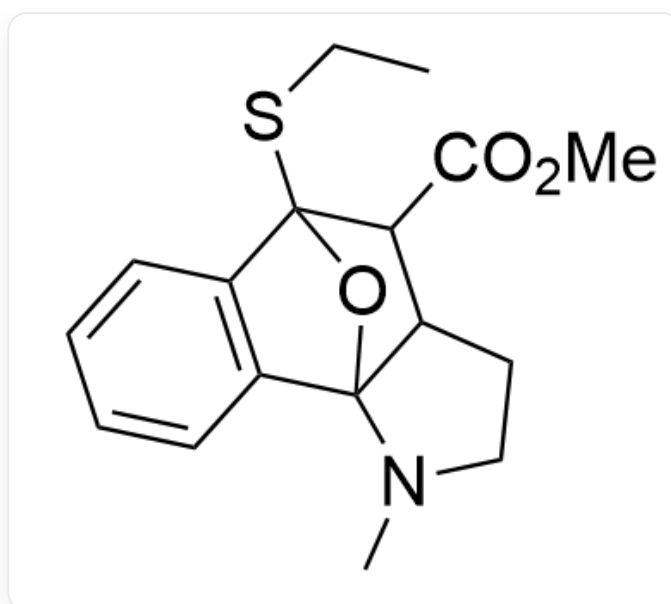
中间体4: CCSC1=C2C(C=CC=C2)=C(O1)N(CC/C=C/C(OC)=O)C

CHECKPOINT

1 PTS

中间体4: CCSC1=C2C(C=CC=C2)=C(O1)N(CC/C=C/C(OC)=O)C

接着发生分子内DA反应，得到中间体5



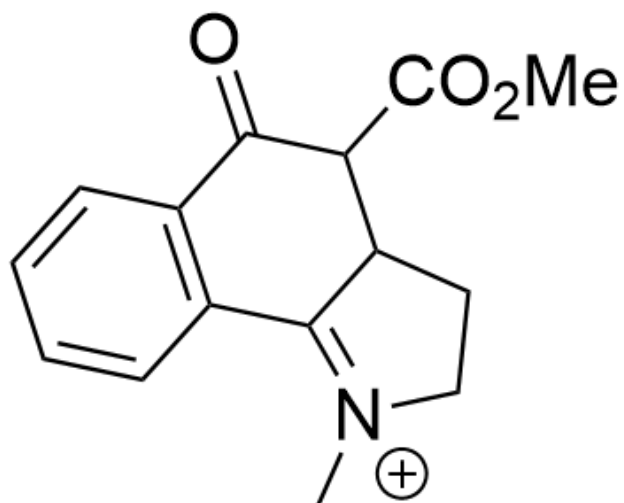
中间体5: CCSC12C3=C(C4(O1)N(CCC4C2C(OC)=O)C)C=CC=C3

CHECKPOINT

1 PTS

中间体5: CCSC12C3=C(C4(O1)N(CCC4C2C(OC)=O)C)C=CC=C3

然后脱去一分子硫醇阴离子得到中间体6



中间体**6**: O=C1C2=C(C3=[N+](CCC3C1C(OC)=O)C)C=CC=C2

CHECKPOINT

1 PTS

中间体**6**: O=C1C2=C(C3=[N+](CCC3C1C(OC)=O)C)C=CC=C2

最后中间体**6**进行芳构化得到产物**A**

